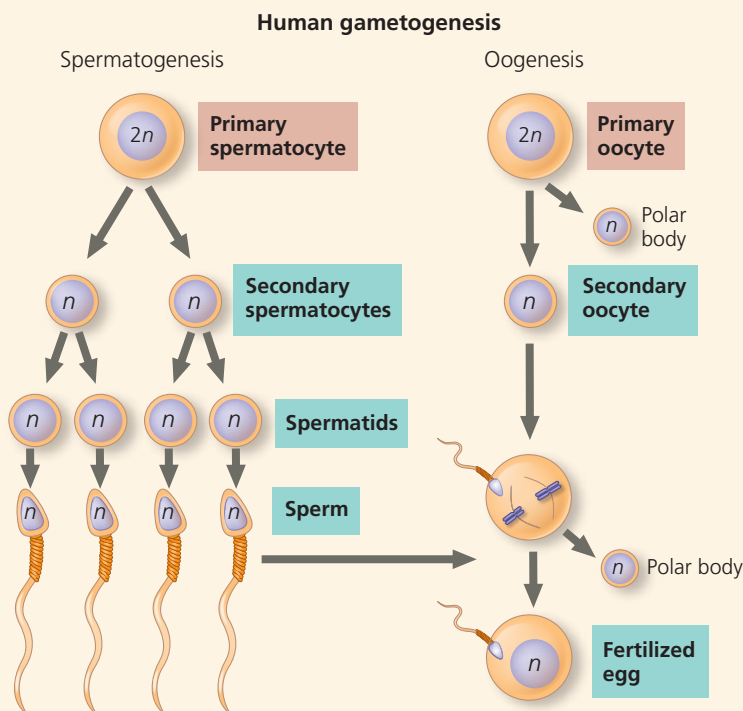


CONCEPT 46.3

Reproductive organs produce and transport gametes (pp. 1025–1029)

- In human males, **sperm** are produced in **testes**, which are suspended outside the body in the **scrotum**. Ducts connect the testes to internal accessory glands and to the **penis**. The reproductive system of the human female consists principally of the **labia** and the **glans** of the **clitoris** externally and the **vagina**, **uterus**, **oviducts**, and **ovaries** internally. **Eggs** are produced in the ovaries and upon fertilization develop in the uterus.
- Gametogenesis**, or gamete production, consists of the processes of **spermatogenesis** in males and **oogenesis** in females. Human spermatogenesis is continuous and produces four sperm per meiosis. Human oogenesis is discontinuous and cyclic, generating one egg per meiosis.

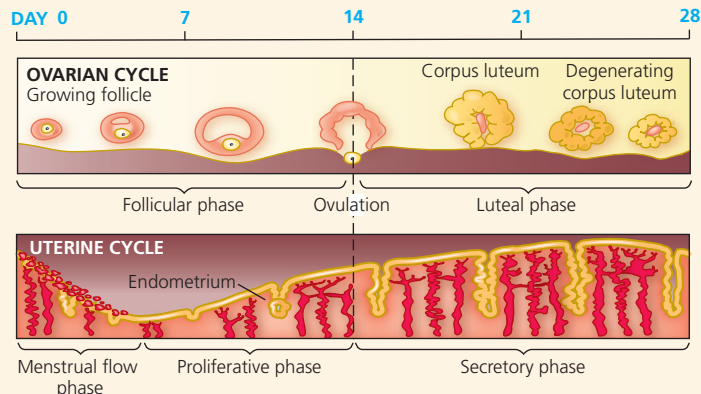


? How does the difference in size and cellular contents between sperm and eggs relate to their specific functions in reproduction?

CONCEPT 46.4

The interplay of tropic and sex hormones regulates reproduction in mammals (pp. 1030–1034)

- Human sexuality, which includes biological sex, gender identity, and sexual orientation, exhibits considerable variation.
- In mammals, GnRH from the hypothalamus regulates the release of two hormones, **FSH** and **LH**, from the anterior pituitary. In males, FSH and LH control the secretion of androgens (chiefly **testosterone**) and sperm production. In females, cyclic secretion of FSH and LH orchestrates the **ovarian** and **uterine cycles** via estrogens (primarily **estradiol**) and **progesterone**. The developing **follicle** and the **corpus luteum** also secrete hormones, which help coordinate the uterine and ovarian cycles through positive and negative feedback.



- In **estrous cycles**, the lining of the **endometrium** is reabsorbed, and sexual receptivity is limited to a heat period. Reproductive structures with a shared origin in development underlie many features of human sexual arousal and orgasm common to males and females.

? Why do anabolic steroids lead to reduced sperm counts?

CONCEPT 46.5

In placental mammals, an embryo develops fully within the mother's uterus (pp. 1034–1040)

- After fertilization and the completion of meiosis in the oviduct, the zygote undergoes a series of cell divisions and develops into a **blastocyst** before implantation in the endometrium. All major organs start developing by 8 weeks. A pregnant woman's acceptance of her "foreign" offspring likely reflects partial suppression of the maternal immune response.
- Contraception** may prevent release of mature gametes from the gonads, fertilization, or embryo implantation. **Abortion** is the termination of a pregnancy in progress.
- Reproductive technologies can help detect problems before birth and can assist infertile couples. Infertility may be treated through hormone therapy or **in vitro fertilization**.

? What route would oxygen in the mother's blood follow to arrive at a body cell of the fetus?

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

- Which of the following characterizes parthenogenesis?
 - An individual may change its sex during its lifetime.
 - Specialized groups of cells grow into new individuals.
 - An organism is first a male and then a female.
 - An egg develops without being fertilized.
- In male mammals, excretory and reproductive systems share
 - the vas deferens.
 - the urethra.
 - the seminal vesicle.
 - the prostate.
- Which of the following is properly paired?
 - seminiferous tubule—cervix
 - vas deferens—oviduct
 - corpus luteum—Sertoli cell
 - scrotum—clitoris